Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel International Award in Lower Secondary

Monday 14 October 2024

Afternoon (Time: 1 hour 20 minutes)

Paper reference **LCP11/01**

Computing

Year 9
Achievement Test



You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- You must **NOT** use a calculator.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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P 7 8 4 0 9 A 0 1 2 4

Pearson

SECTION A

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Computers communicate using networks.

(a) Complete the sentence using the correct word from this list.

- address
- name
- page
- site

A URL is the web ..... that a user enters into a web browser.

(1)

(b) Which **one** of these is a digital device designed to intercept data?

(1)

☒ **A** GPS

☒ **B** Keylogger

☒ **C** Router

☒ **D** WAP

(c) State **one** reason a data packet includes a sequence number.

(1)

.....

(Total for Question 1 = 3 marks)

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2 State what is meant by the term **World Wide Web**.

.....

(Total for Question 2 = 1 mark)

3 Which **one** of these is read by a web browser?

- ☒ **A** HDD
- ☒ **B** HTML
- ☒ **C** HTTP
- ☒ **D** HTTPS

(Total for Question 3 = 1 mark)

4 Complete the sentence using the correct option from this list.

- edit files together
- hide their location
- improve data security
- increase personal privacy

Connecting computers to the Internet allows people to .....

.....

(Total for Question 4 = 1 mark)

5 Complete the sentence using the correct option from this list.

- NFC
- optical
- Wi-Fi
- wireless

A mobile phone network provides ..... connectivity.

(Total for Question 5 = 1 mark)

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6 One type of network is a personal area network.

State **one other** type of network.

.....

(Total for Question 6 = 1 mark)

7 State what is measured in bits per second.

.....

(Total for Question 7 = 1 mark)

8 Which **one** of these is the purpose of a microphone?

- ☒ **A** Input
- ☒ **B** Output
- ☒ **C** Process
- ☒ **D** Storage

(Total for Question 8 = 1 mark)

9 One type of software is proprietary.

State **one other** software type.

.....

(Total for Question 9 = 1 mark)

10 Complete the sentence using the correct word from this list.

- application
- binary
- operating
- utility

A user interface is provided by the ..... system.

(Total for Question 10 = 1 mark)

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11 Computers use storage and memory.

(a) Complete the sentence using the correct option from this list.

- hard-disk storage
- random-access memory
- read-only memory
- solid-state storage

(1)

Data in ..... is lost when a computer is powered off.

(b) Complete the sentence using the correct word from this list.

- connections
- devices
- discs
- instructions

(1)

Main memory stores ..... used by the central processing unit.

(c) Which **one** of these is a reason for using a pen drive?

(1)

- ☒ **A** Better anti-malware
- ☒ **B** Easier to carry around
- ☒ **C** No need to back up
- ☒ **D** Uses magnetic media

(Total for Question 11 = 3 marks)

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12 Draw **one** straight line from **each** type of storage to the device that uses it.

Storage type	Device
	DVD
Magnetic	GPS receiver
Optical	Hard-disk
Solid state	NFC reader
	Pen drive

(Total for Question 12 = 3 marks)

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13 The Python programming language can make use of subprograms.

Here is some Python code.

```
1 sizes = ["S", "M", "XXL"]
2 sizes.append("XL")
3 del sizes[2]
```

(a) Complete the sentence using the correct option from this list.

- an argument
- a parameter
- an operator
- a string

(1)

The code uses the value `"XL"` as ..... for the subprogram.

(b) Which **one** of these is the name of a data structure used in this code?

(1)

☒ **A** `append`

☒ **B** `del`

☒ **C** `sizes`

☒ **D** `XL`

(c) Which **one** of these is the middle item in the list after the code runs?

(1)

☒ **A** M

☒ **B** S

☒ **C** XL

☒ **D** XXL

(Total for Question 13 = 3 marks)

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14 Here is some Python code.

```
1 letter = False
2 word = "tree"
3 "count" = 2
4 print (2)
```

(a) State the number of the line that assigns a string value.

(1)

.....

(b) Explain the error in the code.

(2)

.....

.....

.....

.....

(c) Which **one** of these describes `letter`?

(1)

- ☒ **A** assignment
- ☒ **B** iteration
- ☒ **C** selection
- ☒ **D** variable

(Total for Question 14 = 4 marks)

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15 Complete the output for **each** line of code in the table.

Code	Output
`4 % 2`	
`3 // 2`	

(Total for Question 15 = 2 marks)

16 State **two** ways digital technologies can harm the environment.

1 .....

.....

2 .....

.....

(Total for Question 16 = 2 marks)

17 State **two** responsible ways to respond to inappropriate conduct online.

1 .....

.....

2 .....

.....

(Total for Question 17 = 2 marks)

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18 Computers use binary to represent information.

(a) State the number of bytes in a kibibyte.

(1)

.....

(b) Which **one** of these is the result of this binary addition?

```
  1011
+ 0010
```

(1)

- ☒ **A** 10110010
- ☒ **B** 00101011
- ☒ **C** 1101
- ☒ **D** 1021

(c) State the number of bits in the binary pattern 10110110

(1)

.....

(d) Here is an emoji character.



State the character set required to encode an emoji character.

(1)

.....

(Total for Question 18 = 4 marks)

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QUESTION 19 BEGINS ON THE NEXT PAGE.

19 Here is a list of items.

1, 4, 3, 2, 9, 8

A bubble sort algorithm is used to sort the items in ascending order.

Use this space for working to help you answer the questions on this page.

(a) The first pass is completed.

(i) State the number of swaps made during the first pass.

(1)

.....

(ii) State the value of the item that is now in the last position.

(1)

.....

(b) State the number of passes required to sort the list.

(1)

.....

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(c) Describe the condition that causes a bubble sort algorithm to end.

(2)

.....

.....

.....

.....

(d) A search algorithm is used to find a value in a list.

State **one** type of **search** algorithm.

(1)

.....

(Total for Question 19 = 6 marks)

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20 Data representation is often not perfect.

(a) Describe how increasing bit-depth improves the representation of **sound**.

(2)

(b) Describe how a **bitmap image** is represented in binary.

(3)

(Total for Question 20 = 5 marks)

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21 In a computer game, a player starts with a score of 0

If the player finds a coin, the score increases by 10 and is displayed.

Complete the program by adding Python code on the dotted lines.

```
1 score = 0
2 if coin_found():
3       ..... = ..... + .....
4       print("Your score is: " ..... score)
```

(Total for Question 21 = 4 marks)

TOTAL FOR SECTION A = 50 MARKS

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

22 Databases are used to organise data.

(a) Which **one** of these is needed to link tables in a database?

(1)

- ☒ **A** Operators
- ☒ **B** Primary keys
- ☒ **C** Queries
- ☒ **D** Records

(b) Which **one** of these describes a foreign key?

(1)

- ☒ **A** Only used in online databases
- ☒ **B** Only used in queries
- ☒ **C** Refers to a key in another table
- ☒ **D** Uses the name of a report

(c) Which **one** of these is a logical operator used in a database query?

(1)

- ☒ **A** AVG
- ☒ **B** DIV
- ☒ **C** LIKE
- ☒ **D** NOT

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(d) Complete the sentence using the correct word from this list.

- Boolean
- Date
- Number
- Text

(1)

In a database, the ..... data type has one of two possible values.

(Total for Question 22 = 4 marks)

23 Complete the sentence using the correct word from this list.

- application
- brightness
- licence
- utility

Images can be edited by adjusting the ..... and contrast.

(Total for Question 23 = 1 mark)

24 Which **one** of these is used to change the dimensions of an image?

- ☒ **A** Compress
- ☒ **B** Erase
- ☒ **C** Recolour
- ☒ **D** Resize

(Total for Question 24 = 1 mark)

25 State **one** reason for compressing a file before sharing it online.

.....

(Total for Question 25 = 1 mark)

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26 State **one** way a vector graphic is different from a bitmap graphic.

.....

(Total for Question 26 = 1 mark)

27 Images are often included in presentations.

(a) Complete the sentence using the correct word from this list.

- anchor
- multimedia
- text
- transition

(1)

In presentation software, embedding ..... content, such as video, enhances the user experience.

(b) Complete the sentence using the correct word from this list.

- bookmarks
- functions
- images
- transitions

(1)

A presentation aimed at young children should use ..... rather than text.

(Total for Question 27 = 2 marks)

28 Which **one** of these controls access to files?

☒ **A** Abstraction

☒ **B** Collaboration

☒ **C** Formatting

☒ **D** Permissions

(Total for Question 28 = 1 mark)

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29 Which **one** of these is a cloud-based service that provides access to software over the Internet?

- ☒ **A** Disk utility
- ☒ **B** File handling
- ☒ **C** Hosted applications
- ☒ **D** Web filter

(Total for Question 29 = 1 mark)

30 State **one** reason for using page breaks in documents.

.....

(Total for Question 30 = 1 mark)

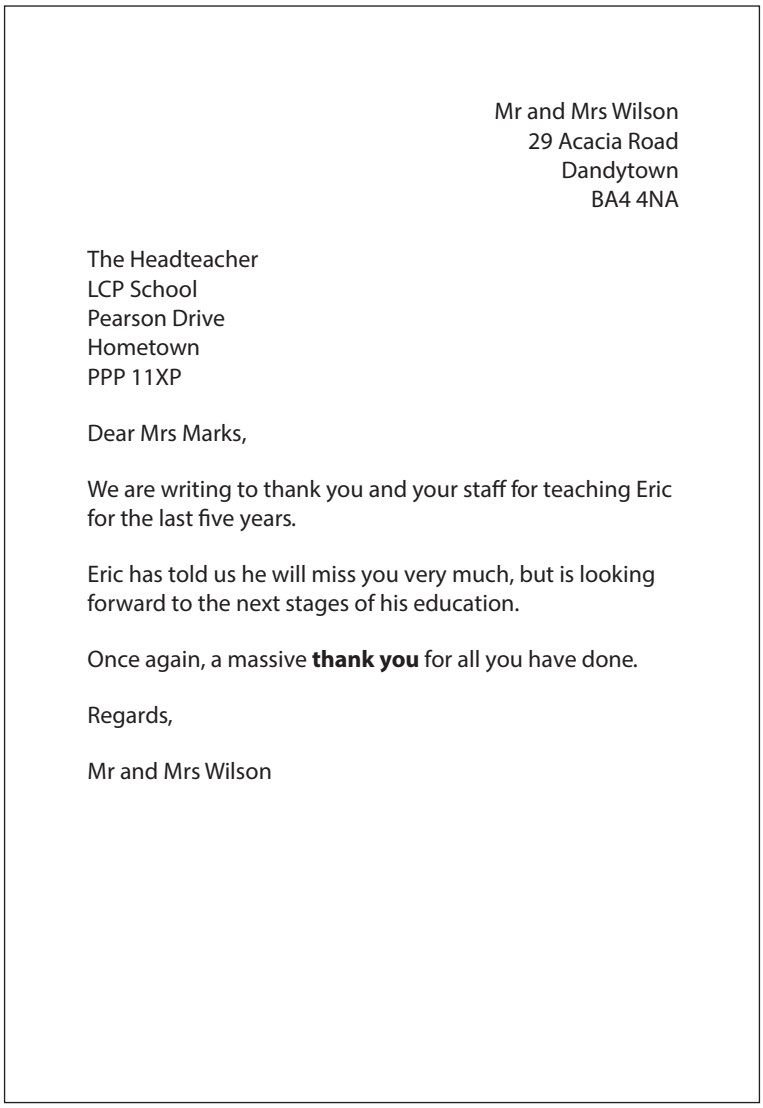
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31 Look at **Figure 1**.

It shows a letter.



Mr and Mrs Wilson
29 Acacia Road
Dandytown
BA4 4NA

The Headteacher
LCP School
Pearson Drive
Hometown
PPP 11XP

Dear Mrs Marks,

We are writing to thank you and your staff for teaching Eric for the last five years.

Eric has told us he will miss you very much, but is looking forward to the next stages of his education.

Once again, a massive **thank you** for all you have done.

Regards,

Mr and Mrs Wilson

Figure 1

(a) State **two** formatting options used in the letter.

(2)

1 .....

2 .....

(b) State **one** way a memo differs from a letter.

(1)

.....

(Total for Question 31 = 3 marks)

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32 Look at **Figure 2**.

It shows part of a spreadsheet used by a school library.

	A	B	C
1	Books to read this month:	30	
2			
3	**Student**	**Books read**	**% of books read**
4	A	12	40%
5	B	9	30%
6	C	20	67%
7	D	23	77%
8	E	21	70%

Figure 2

(a) State the number formatting option applied to cell C4.

(1)

.....

(b) The library wants to compare the number of books each student has read.

State the type of chart needed to show this.

(1)

.....

(c) The number of books on the reading list changes monthly.

Write the formula that is used in cell C8.

(3)

(d) The library tracks students' monthly reading.

Explain **one** reason multiple worksheets are used for this purpose.

(2)

.....

.....

.....

.....

(Total for Question 32 = 7 marks)

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33 A company report uses a house style.

Describe what is meant by **house style**.

.....

.....

.....

.....

(Total for Question 33 = 2 marks)

34 Explain **one** reason why it is important to gather information from a range of sources before using it in a presentation.

.....

.....

.....

.....

(Total for Question 34 = 2 marks)

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35 Look at **Figure 3**.

It shows part of a web page.

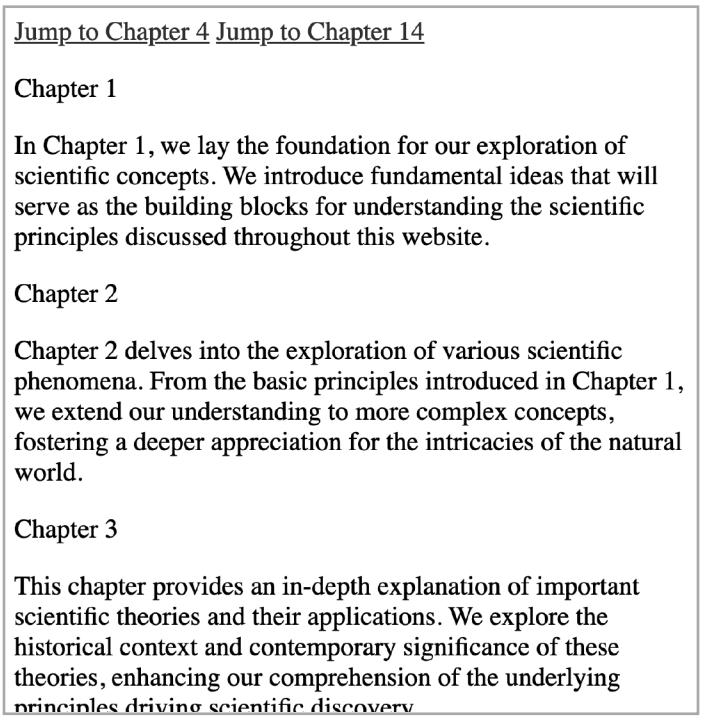


Figure 3

(a) There are two hyperlink anchors at the top of the web page.

State the type of HTML element they link to.

(1)

.....

(b) One way that HTML can be used to improve the appearance of the web page is to use `<br>` elements to break up blocks of text.

Describe **one other** way HTML can be used to improve the appearance of the web page.

(2)

.....

.....

.....

.....

(Total for Question 35 = 3 marks)

TOTAL FOR SECTION B = 30 MARKS
TOTAL FOR PAPER = 80 MARKS

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